Akoizo Original Practice - Form 1

Division C | Boomilever | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) A structure weighs 8.00 g and holds 15,000 g without the bonus. What is its score?

A. 1,875

B. 120

C. 15,000

D. 1,500

Answer: _____

2. (2 points) How may participants keep the bucket from swinging during loading?

A. By resting it on a stool

B. By pushing the structure

C. Only with the tips of the stabilizing sticks

D. By holding the bucket handle

Answer: _____

3. (2 points) Doubling a strut’s unbraced length changes its Euler buckling load by a factor of:

A. 2

B. 4

C. 1/4

D. 1/2

Answer: _____

4. (4 points) Calculate the moment of a 147 N load hung 0.44 m from the wall.

Answer: _____
_____
_____

5. (2 points) Why change only one variable between test builds?

Unit: Testing, failure analysis, and iteration.
Worked-case data: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?
Separate investigation observations:
Builds 1-3: 8.1 g, 8.3 g, 8.2 g: all held 15 kg.
Builds 4-5: 7.4 g, 7.5 g: failed at 12 kg and 14 kg, strut buckled.
Build 6: 7.6 g with a stiffer, denser strut: held 15 kg.

A. So any change in result can be traced to that variable

B. It is required by the rules

C. It saves glue

D. It makes the structure lighter

Answer: _____

6. (4 points) Why are adhesive tapes and glue-soaked wood both banned, and what does that mean for design?

Unit: Joints, adhesives, and the bolt hook.
Worked-case data: A chord joint must carry 300 N. Your glue holds about 2 N per mm^2 of long-grain overlap. How much bonded area do you need, with a safety factor of 1.5?
Separate investigation observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

Answer: _____
_____
_____

7. (4 points) What does a pyramid layout gain over a single flat triangle?

Unit: Geometry: contact depth, strut angle, and the bonus line.
Worked-case data: Compare strut forces for a full 15 kg (147 N) at 42 cm reach, with contact at 15 cm versus 10 cm.
Separate investigation observations:
Design X: Reach 45 cm, contact 14 cm: strut needed 2 braces.
Design Y: Reach 41 cm, contact 14 cm: strut needed 1 brace.
Design Z: Reach 41 cm, contact 9 cm: strut needed 3 braces.

Answer: _____
_____
_____

8. (2 points) Define “Load Scored” in the context of Boomilever.

Answer: _____
_____
_____

9. (4 points) List three situations that end loading.

Unit: The test apparatus, setup, and loading procedure.
Worked-case data: Plan the six-minute window for a team that has practiced setup.
Separate investigation observations:
Run 1: Setup took 5 min; fast pour; failed at 9 kg with violent swing.
Run 2: Setup took 2 min; slow pour with stick tips; held 15 kg.
Inspection: Both structures were identical builds from the same wood batch.

Answer: _____
_____
_____

10. (4 points) Can a team glue three balsa strips into one thicker strut? Explain.

Answer: _____
_____
_____

11. (2 points) Define “Moment” in the context of Boomilever.

Answer: _____
_____
_____

12. (6 points) Solve this case and explain your reasoning: Your log shows: 9.2 g held 15 kg; 8.4 g held 15 kg; 7.9 g failed at 13.1 kg (strut buckled). What should the competition build be?

Answer: _____
_____
_____

13. (2 points) Why did the fix work?

Supplied synthetic observations:
Failure 1: Hook crushed into an oval around the bolt at 10 kg.
Failure 2: Strut-to-chord butt joint split at 13 kg.
Fix trial: Basswood bearing pad (+0.15 g) and two gussets (+0.10 g): held 15 kg.

A. It lowered bearing stress at the bolt and increased joint glue area

B. It made the strut longer

C. It moved the chain farther from the wall

Answer: _____

14. (2 points) Define “Strut angle” in the context of Boomilever.

Answer: _____
_____
_____

15. (2 points) Define “Contact depth line” in the context of Boomilever.

Answer: _____
_____
_____

16. (2 points) Define “Contact width lines” in the context of Boomilever.

Answer: _____
_____
_____

17. (2 points) Define “Butt joint” in the context of Boomilever.

Answer: _____
_____
_____

18. (2 points) Define “Unbraced length” in the context of Boomilever.

Answer: _____
_____
_____